From the Geometric Interpretation of Berry's Phase to Chern Numbers and Bulk–Edge Correspondence

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1 Basic concepts in geometry

The geometric background in this section can be found in Lee's books on smooth manifolds and Riemannian manifolds [1, 2]. We work with smooth manifolds throughout.

1.1 The tangent space

Here we will introduce the tangent space on a general real manifold.

1.1.1 What is a manifold?

Roughly speaking, an n-dimensional manifold is something that locally looks like $\mathbb{R}^n$. For example, the 2-sphere $S^2 \subset \mathbb{R}^3$ is a two-dimensional manifold. But the figure-eight is not a manifold (any neighborhood of the intersection point doesn't look like $\mathbb{R}$). Manifolds are often introduced as subsets of some $\mathbb{R}^N$, but an abstract manifold does not come with a chosen embedding into Euclidean space. For example, the Brillouin zone for a 2D square lattice is a manifold, but we need not think of this Brillouin zone as living in some $\mathbb{R}^n$ (although it can).

1.1.2 Tangent vectors of $S^2 \subset \mathbb{R}^3$ and $\mathbb{R}^3$

Before introducing abstract tangent vectors, we go back and take a closer look at the usual tangent vectors. Let $p \in S^2$ be a point on the unit sphere. We say a vector v is tangent to S^2 at p if there is a curve $\gamma : (-\epsilon, \epsilon) \to S^2$ such that $\gamma(0) = p$ and $\gamma'(0) = v$. A similar definition can be made for tangent vectors in $\mathbb{R}^3$.

1.1.3 Tangent vectors as derivations and the tangent space

A tangent vector in $\mathbb{R}^3$ can be viewed as a directional derivative in the following way. Given $p \in \mathbb{R}^3$ and a tangent vector v tangent to $\mathbb{R}^3$ at p, we can find a curve $\gamma : (-\epsilon, \epsilon) \to \mathbb{R}^3$ such that $\gamma(0) = p$ and $\gamma'(0) = v$. Define an operation D_v from the space of real-valued smooth functions on $\mathbb{R}^3$ to $\mathbb{R}$ in the following way

$$D_v(f) \equiv \left.\frac{df(\gamma(t))}{dt}\right|_{t=0} \tag{1}$$

In other words, D_v takes the directional derivative along v at p. The essential algebraic properties of D_v are first, D_v is linear in f

$$D_v(\alpha f + \beta g) = \left.\frac{d[\alpha f(\gamma(t)) + \beta g(\gamma(t))]}{dt}\right|_{t=0} = \alpha D_v(f) + \beta D_v(g). \tag{2}$$

And second, D_v satisfies the Leibniz rule

$$D_v(fg) = \left.\frac{d[f(\gamma(t))g(\gamma(t))]}{dt}\right|_{t=0} = g(p)D_v(f) + f(p)D_v(g). \tag{3}$$

We define tangent vectors on general manifolds to be operations with these two properties.

Definition 1.1. *Let M be a smooth manifold, $p \in M$ be a point on M. A real-linear map $D : C^\infty(M, \mathbb{R}) \to \mathbb{R}$ is called a tangent vector on M at p if it is linear and satisfies the Leibniz rule, i.e., if*

$$\begin{aligned} D(\alpha f + \beta g) &= \alpha D(f) + \beta D(g) \\ D(fg) &= f(p)D(g) + g(p)D(f). \end{aligned} \tag{4}$$

The set of all tangent vectors of M at p is denoted as T_pM. It is worth noting that T_pM is a vector space.

1.1.4 A basis of the tangent space

Suppose M is an n-dimensional manifold and $p \in M$ is a point. Since M locally looks like $\mathbb{R}^n$, we can find an open subset U of M containing p and a coordinate map $\phi : U \to \phi(U) \subset \mathbb{R}^n$ onto an open subset, which is a diffeomorphism (a diffeomorphism is a smooth map that is one to one and onto, whose inverse is also smooth), $\phi(q) = (x^1(q), ..., x^n(q))$, $\forall q \in U$. Define $\frac{\partial}{\partial x^k}\big|_p$ as the operation

$$\left.\frac{\partial}{\partial x^k}\right|_p (f) \equiv \left.\frac{\partial f(\phi^{-1}(y^1, \ldots, y^n))}{\partial y^k}\right|_{(y^1,\ldots,y^n)=\phi(p)} \tag{5}$$

One can prove that $\{\frac{\partial}{\partial x^1}\big|_p, ..., \frac{\partial}{\partial x^n}\big|_p\}$ is a basis of $T_p(M)$. Therefore, the tangent space of a manifold of dimension n is also of dimension n.

1.2 Differential forms

Here we introduce a very important concept in geometry: differential forms. We first use real-valued forms; allowing complex coefficients gives the complex-valued forms needed for Berry's connection. We will use differential forms to represent geometric quantities such as connections and curvature. Differential forms can also be used to represent topological invariants. In particular, we will use them to construct the Chern classes, which give us Chern numbers by integration.

1.2.1 Formal definition of differential forms

Definition 1.2. *A differential k-form at $p \in M$ is an alternating multilinear map*

$$\begin{aligned} \omega_p : T_pM \times \cdots \times T_pM &\to \mathbb{R} \\ (v_1, ..., v_k) &\mapsto \omega_p(v_1, ..., v_k) \end{aligned} \tag{6}$$

In this definition, linear means that ω_p is linear in each argument separately. Alternating means

$$\omega_p(..., v_i, ..., v_j, ...) = -\omega_p(..., v_j, ..., v_i, ...). \tag{7}$$

It is worth noting that the differential forms at $p \in M$ form a vector space. A differential form on M is obtained by gluing together differential forms at each point in a smooth way.

Definition 1.3. *A differential k-form ω on M is an assignment of ω_p at each point $p \in M$ in a smooth way: for all smooth vector fields $v_1, ..., v_k$, the function mapping p to $\omega_p(v_1|_p, ..., v_k|_p)$ is a smooth function. In particular, we define a smooth function on M to be a differential 0-form.*

It is worth noting that we can multiply a differential k-form by a smooth function. Just let $(f\omega)_p \equiv f(p)\omega_p$.

1.2.2 1-form: the cotangent vector

A 1-form at $p \in M$, which is also called a cotangent vector at $p \in M$, is simply a linear map from T_pM to $\mathbb{R}$. The set of 1-forms at p forms a vector space T_p^*M, called the cotangent space. Let U be an open neighborhood of p and $\phi : U \to \mathbb{R}^n$ be a coordinate chart of M, where $\phi(q) = (x^1(q), ..., x^n(q))$. Then $\{\frac{\partial}{\partial x^1}\big|_p, ..., \frac{\partial}{\partial x^n}\big|_p\}$ is a basis of $T_p(M)$. Let $dx^i\big|_p$ be the linear map of T_pM such that $dx^i\big|_p(\frac{\partial}{\partial x^j}\big|_p) = \delta^i_j$. Then the set $\{dx^1\big|_p, ..., dx^n\big|_p\}$ forms a basis of the cotangent space. Gluing $dx^i\big|_p$ together on a coordinate neighborhood, we obtain a differential form dx^i on that neighborhood. Locally, a generic 1-form can be written as $\omega = \sum_{i=1}^n f_i dx^i$.

1.2.3 Building higher forms from 1-forms: the wedge product

Now we build k-forms from 1-forms using the wedge product.

Definition 1.4. *Let $\omega_1, ..., \omega_k$ be k 1-forms. The wedge product $\omega_1 \wedge \cdots \wedge \omega_k$ is defined as a map*

$$(\omega_1 \wedge \cdots \wedge \omega_k)(v_1, ..., v_k) = \det \begin{pmatrix} \omega_1(v_1) & \cdots & \omega_1(v_k) \\ \vdots & \ddots & \vdots \\ \omega_k(v_1) & \cdots & \omega_k(v_k) \end{pmatrix} \tag{8}$$

where $v_1, ..., v_k$ are smooth vector fields.

Note that Eq. (8) is multilinear and alternating in its vector arguments, so it defines a differential k-form. An important property of the wedge product is that $\omega_1 \wedge \omega_2 = -\omega_2 \wedge \omega_1$ for 1-forms ω_1 and ω_2.

1.2.4 k-forms in local coordinates

Proposition 1.1. *Let ω_p be a differential k-form at $p \in M$. Then*

$$\omega_p = \sum_{i_1 < ... < i_k} a_{i_1,...,i_k} (dx^{i_1} \wedge \cdots \wedge dx^{i_k})|_p, \tag{9}$$

where $a_{i_1,...,i_k} \in \mathbb{R}$ are coefficients.

By Proposition 1.1, a generic k-form can be written locally as

$$\omega = \sum_{i_1 < ... < i_k} f_{i_1,...,i_k} dx^{i_1} \wedge \cdots \wedge dx^{i_k} \tag{10}$$

where $f_{i_1,...,i_k}$ are smooth functions on the coordinate neighborhood.

1.2.5 The wedge product for general forms

Let $\omega = dx^{i_1} \wedge \cdots \wedge dx^{i_k}$ be a k-form and $\eta = dx^{j_1} \wedge \cdots \wedge dx^{j_l}$ be an l-form. The wedge product of ω and η is defined as

$$\omega \wedge \eta \equiv dx^{i_1} \wedge \cdots \wedge dx^{i_k} \wedge dx^{j_1} \wedge \cdots \wedge dx^{j_l}. \tag{11}$$

We extend this definition by bilinearity to arbitrary forms. For a k-form ω and an l-form η, we have $\omega \wedge \eta = (-1)^{kl}\eta \wedge \omega$. From now on, we also allow complex-valued forms by allowing their coefficients to lie in $\mathbb{C}$.

2 Geometric interpretation of Berry's phase, Berry's connection and Berry's curvature

2.1 The physical setup

Here we talk about the physical setup. Consider a tight-binding model of non-interacting spinless fermions on a 2D Bravais lattice $\Gamma = \{x\vec{r}_1 + y\vec{r}_2 \mid x, y \in \mathbb{Z}\}$ with n degrees of freedom per unit cell. We use fermions so that filling all bands below a Fermi level gives a band insulator. The geometry of an isolated group of single-particle bands can also be studied for number-conserving bosonic models, but their many-particle ground states need not fill those bands [3]. Let $c^\dagger_{i,x,y}$ create a fermion of type i in the cell $x\vec{r}_1 + y\vec{r}_2$. The canonical anticommutation relations include

$$\{c_{i,w,x}, c^\dagger_{j,y,z}\} = \delta_{ij}\delta_{wy}\delta_{xz}. \tag{12}$$

To keep the normalization explicit, first take a finite periodic lattice with N unit cells and define

$$c_i(\vec{k})^\dagger = \frac{1}{\sqrt{N}} \sum_{x,y} c^\dagger_{i,x,y} e^{i\vec{k}\cdot(x\vec{r}_1 + y\vec{r}_2)}. \tag{13}$$

The allowed momenta are discrete. In the infinite-lattice limit they fill the Brillouin zone $BZ = \mathbb{R}^2/\Gamma^*$, where Γ^* is the reciprocal lattice. Thus BZ is a two-dimensional torus. A translation-invariant, particle-number-conserving quadratic Hamiltonian can be written as

$$\widehat{H} = \sum_{\vec{k}} \sum_{i,j=1}^{n} H_{ij}(\vec{k}) c_i(\vec{k})^\dagger c_j(\vec{k}), \qquad H(\vec{k}) = H(\vec{k})^\dagger. \tag{14}$$

Here $H(\vec{k})$ denotes the $n \times n$ single-particle Bloch Hamiltonian. We assume it is smooth and periodic; finite-range hopping is one sufficient condition. We assume a gap at the Fermi energy $\epsilon = 0$, with

$$\lambda_1(\vec{k}) \le \cdots \le \lambda_m(\vec{k}) < 0 < \lambda_{m+1}(\vec{k}) \le \cdots \le \lambda_n(\vec{k}). \tag{15}$$

Choose the eigenvectors to form an orthonormal basis, including within each degenerate eigenspace. Write the eigenvector associated with $\lambda_i(\vec{k})$ as

$$\psi_i(\vec{k}) = \sum_{l=1}^{n} a_{il}(\vec{k}) c_l(\vec{k})^\dagger |0\rangle, \qquad \sum_{l=1}^{n} |a_{il}(\vec{k})|^2 = 1. \tag{16}$$

Eigenvectors need only be chosen locally. At a degeneracy, it is the whole occupied subspace, rather than each eigenvector separately, that varies smoothly. Let the Bloch state associated with the jth degree of freedom be

$$\phi_j(\vec{k}) = c_j(\vec{k})^\dagger |0\rangle. \tag{17}$$

We will show next that these orbital Bloch states provide a global frame for a trivial vector bundle, while $H(\vec{k})$ can define a nontrivial subbundle.

2.2 The geometry of the trivial Hermitian bundle E formed by all bands

Here we show that the set of all bands forms a trivial bundle and hence is topologically trivial as a bundle.

2.2.1 The trivial Hermitian bundle E of a physical system on a lattice

Roughly speaking, we obtain a Hermitian vector bundle over a manifold M by putting a complex inner-product space $V(p)$ of fixed dimension n at each point $p \in M$ and gluing these spaces together smoothly. A bundle is trivial if it is isomorphic to the product bundle $M \times \mathbb{C}^n$. For a familiar real-vector-bundle analogy, $S^1 \times \mathbb{R}$ is a trivial real line bundle, whereas the open Möbius strip is a nontrivial real line bundle over S^1. These examples have real, rather than complex, one-dimensional fibers. Now we define the trivial Hermitian bundle E of our lattice system. Its base manifold is the Brillouin zone BZ, and its fiber at $\vec{k}$ is $V(\vec{k}) \equiv \text{Span}\{\phi_1(\vec{k}), \ldots, \phi_n(\vec{k})\}$. On the finite periodic lattice, the anticommutation relations give

$$\{c_i(\vec{k}_1), c_j(\vec{k}_2)^\dagger\} = \delta_{ij}\delta_{\vec{k}_1,\vec{k}_2}, \qquad \langle \phi_i(\vec{k}_1), \phi_j(\vec{k}_2)\rangle = \delta_{ij}\delta_{\vec{k}_1,\vec{k}_2}. \tag{18}$$

In the infinite-lattice limit, Bloch waves are generalized states, and the full-space inner product involves a delta function. The finite-dimensional Bloch fiber nevertheless has the ordinary inner product

$$\langle \phi_i(\vec{k}), \phi_j(\vec{k})\rangle = \delta_{ij}. \tag{19}$$

We use the convention that the inner product is conjugate-linear in its first argument.

The vector spaces are glued in a way such that $\sum_{j=1}^n a_j\phi_j(\vec{k}_1)$ and $\sum_{j=1}^n c_j\phi_j(\vec{k}_2)$ are close to each other if and only if $\vec{k}_1$ and $\vec{k}_2$, a_1 and c_1,..., a_n and c_n are all close to each other. This Hermitian bundle E is trivial because the map

$$\begin{aligned} \eta : E &\to BZ \times \mathbb{C}^n \\ \sum_i a_i\phi_i(\vec{k}) &\mapsto (\vec{k}, (a_1, \ldots, a_n)), \end{aligned} \tag{20}$$

is a smooth bundle isomorphism between E and $BZ \times \mathbb{C}^n$. In particular, it is a homeomorphism of the total spaces. (A homeomorphism is a continuous map which is one to one and onto, whose inverse is also continuous. Two topological spaces are thought to be equivalent if there is a homeomorphism between them.) Since E is a trivial bundle, the Chern number associated to it must be zero. Translating into the language of physics, the first Chern number of the full set of bands must be zero. If the bands are individually isolated, this is the sum of their individual Chern numbers.

2.2.2 The geometric picture of a connection on a Riemannian manifold

Before defining connections on vector bundles, we discuss them on 2D Riemannian manifolds to gain a geometric picture. A 2D Riemannian manifold M is a 2D manifold equipped with a smoothly varying positive-definite inner product g on its tangent spaces, called a metric tensor. By convention, we write $g(X, Y)$ as $\langle X, Y\rangle$. A geodesic has constant speed and is locally length-minimizing: a sufficiently short segment minimizes length between its endpoints. Its length is $L(\gamma) = \int_0^r \sqrt{\langle \gamma'(t), \gamma'(t)\rangle}\, dt$. A long geodesic need not be a shortest path between its endpoints. We should think of the tangent vector $\gamma'(t)$ as not changing its length and direction (if this is confusing, think of a line in $\mathbb{R}^3$). Let v be a smooth vector field on M, $p \in M$ a point on M and $X_p \in T_pM$ a tangent vector of M at p. To define the rate of change of v at p in the direction X_p, we can do the following. Let $\gamma : [0, r] \to M$ be a geodesic starting from p with initial vector $\gamma'(0) = X_p$. Consider the point $\gamma(dt)$ and the vector $v(\gamma(dt))$ on it. Move $v(\gamma(dt))$ back from the point $\gamma(dt)$ to $\gamma(0)$ along the curve γ in a way such that its length and its oriented angle with $\gamma'(t)$ do not change during this motion. Let the resulting vector in T_pM be $w(dt)$. The covariant derivative of v at p along the direction X_p is defined as

$$\nabla_{X_p} v \equiv \lim_{dt\to 0} \frac{w(dt) - v(\gamma(0))}{dt}. \tag{21}$$

From the definition we know that for any smooth function f on M and smooth vector fields X and v on M, we have $\nabla_{fX} v = f\nabla_X v$ and $\nabla_X(fv) = f\nabla_X v + X(f)v$. These two equations will be the key ingredients in defining the connection on a vector bundle.

A side note is that this connection is the Levi–Civita connection: it is the unique connection that is both compatible with the metric and torsion-free [2, Chapter 5]. Metric compatibility means $X\langle u, v\rangle = \langle \nabla_X u, v\rangle + \langle u, \nabla_X v\rangle$, and being torsion-free means $\nabla_X Y - \nabla_Y X = [X, Y]$. Metric compatibility alone does not give uniqueness.

2.2.3 Comparing neighboring vectors: the connection ∇ on E

Technically speaking, if $\vec{k}_1 \neq \vec{k}_2$, $\psi_i(\vec{k}_1)$ and $\psi_i(\vec{k}_2)$ belong to different vector spaces. Elements in different vector spaces cannot be added or subtracted. Then what do we mean by the notation $\frac{\partial \psi_i}{\partial k^1}$? In differential geometry, if we want to compare vectors in nearby vector spaces, we need the concept of connection. Therefore, notations like $\frac{\partial \psi_i}{\partial k^1}$ must correspond to a choice of connection on the vector bundle E. What connection are we using in

physics? To answer this question, we first introduce a connection in differential geometry.
Let $\mathfrak{X}(M)$ be the set of all smooth vector fields on M, $\Gamma(E)$ be the set of all smooth sections of E. Here by a smooth section of E we mean at each point $\vec{k} \in M$, we choose a vector $s(\vec{k}) \in V(\vec{k})$ and make this choice smoothly. Following our discussion of connection on a Riemannian manifold, we have the following definition for a connection on vector bundles.

Definition 2.1. *A connection on a vector bundle E with base manifold M is a map*

$$\begin{aligned} \nabla : \mathfrak{X}(M) \times \Gamma(E) &\to \Gamma(E) \\ (X, s) &\mapsto \nabla_X s \end{aligned} \tag{22}$$

which is additive in both arguments and complex-linear in the section argument, and satisfies, for smooth functions f and g on M (with f real-valued),

$$\nabla_{fX}(gs) = fg\nabla_X s + fX(g)s \tag{23}$$

In particular, we see from the definition that ∇ is $C^\infty(M, \mathbb{R})$-linear in X, i.e., $\nabla_{fX} s = f\nabla_X s$ for all real-valued smooth functions f on M. ∇ satisfies the Leibniz rule in s, i.e., $\nabla_X(gs) = g\nabla_X s + X(g)s$.

Definition 2.2. *A section s is said to be parallel (or flat) if $\nabla_X s = 0$ for all smooth vector fields X.*

If a frame is required to consist of parallel sections, this requirement uniquely determines a connection. A general connection need not admit such a frame. For the bundle E, if we require that the sections corresponding to the orbital Bloch states $\phi_i(\vec{k})$ are all parallel, i.e., $(\nabla_X \phi_i)(\vec{k}) = 0$ for all smooth vector fields X, then ∇ is uniquely determined. This is because any vector field can be written locally as $X(\vec{k}) = \sum_{i=1}^2 X^i(\vec{k}) \frac{\partial}{\partial k^i}\big|_{\vec{k}}$ and any section can be written locally as $s(\vec{k}) = \sum_{j=1}^n a_j(\vec{k})\phi_j(\vec{k})$. Using the defining properties of ∇, we obtain

$$(\nabla_X s)(\vec{k}) = \sum_{j=1}^n X(a_j)(\vec{k})\phi_j(\vec{k}) = \sum_{j=1}^n \sum_{i=1}^2 X^i(\vec{k}) \frac{\partial a_j}{\partial k^i}(\vec{k})\phi_j(\vec{k}). \tag{24}$$

Eq. (24) defines a connection on the trivial bundle E and recovers the usual notion of taking derivatives in the computation of Berry's phase.

2.3 Where is the nontrivial topology?

The bundle E formed by all bands is a trivial bundle. But we didn't use any information about the Hamiltonian $H(\vec{k})$ in the construction of E. Here, we show that $H(\vec{k})$ cuts out a possibly nontrivial subbundle of E.

2.3.1 Example: $S^2 \subset \mathbb{R}^3$

$\mathbb{R}^3$ is the simplest 3D manifold and we usually think of $\mathbb{R}^3$ as having trivial topology. However, $\mathbb{R}^3$ contains many subsets that have nontrivial topology. For example, $S^2 \subset \mathbb{R}^3$ is a subset of $\mathbb{R}^3$ and it has nontrivial topology (it has a nontrivial second homotopy group). We obtain S^2 by cutting $\mathbb{R}^3$ using the equation $x^2 + y^2 + z^2 = 1$. This idea extends to vector bundles.

2.3.2 The lowest m bands define a possibly nontrivial subbundle F of E

At each $\vec{k} \in BZ$ we have a vector space $V(\vec{k})$. Since we usually are only interested in the topology of the occupied states, we focus on the bands below the Fermi level $\epsilon = 0$. In this case, at each $\vec{k} \in BZ$, we obtain a subspace $W(\vec{k}) \equiv \text{Span}\{\psi_1(\vec{k}), ..., \psi_m(\vec{k})\} \subset V(\vec{k})$. The subspace $W(\vec{k})$ is defined implicitly by $H(\vec{k})$ and can be thought of as cut out from $V(\vec{k})$ by $H(\vec{k})$. All $W(\vec{k})$ together form a subbundle F of E and this bundle can have nontrivial topology.

2.3.3 Comment on the energy gap

The energy gap has both a geometric and a physical role. Geometrically, it ensures that the occupied spectral projection $P(\vec{k})$ has constant rank and depends smoothly on $\vec{k}$, so its image forms a smooth vector bundle. If the gap closes, this construction can fail. Some gapless systems still have a well-defined smooth subbundle, but this needs a separate argument. Physically, a gap separating the chosen bands from the others is also the hypothesis that makes adiabatic evolution within that group of bands possible [4].

2.4 Berry's connection is the induced connection

The Hermitian bundle E has a connection ∇, and F is a subbundle of E. It is natural to ask if F can inherit these two structures from E. The inner product on F is easy: since $W(\vec{k})$ is a subspace of $V(\vec{k})$, it naturally inherits an inner product from E. But what about the connection? Suppose that, for a section s of F and a smooth vector field $X \in \mathfrak{X}(BZ)$, we naively define $\nabla^F_X s \equiv \nabla_X s$. Then the problem is that $(\nabla_X s)(\vec{k})$ may not be in the subspace $W(\vec{k})$.

2.4.1 The induced connection ∇^F

To deal with the problem that $(\nabla_X s)(\vec{k})$ may not be in the subspace $W(\vec{k})$, we need to find some clever way to put $(\nabla_X s)(\vec{k})$ into $W(\vec{k})$. The key observation is that $V(\vec{k})$ is a Hilbert space. For a Hilbert space, we can define the orthogonal projection $P(\vec{k})$ from $V(\vec{k})$ to $W(\vec{k})$. To be precise,

$$P(\vec{k})(v(\vec{k})) = \sum_{i=1}^{m} \langle \psi_i(\vec{k}), v(\vec{k}) \rangle \psi_i(\vec{k}),\ \forall v(\vec{k}) \in V(\vec{k}). \tag{25}$$

With $P(\vec{k})$, we define

$$(\nabla^F_X s)(\vec{k}) \equiv P(\vec{k})[(\nabla_X s)(\vec{k})] \tag{26}$$

Proposition 2.1. *∇^F defined in Eq. (26) is a connection on F.*

The proof is easy: check additivity and Eq. (23). Since we obtain ∇^F in a natural way from ∇, it is very tempting to say that ∇^F is Berry's connection. But in what sense? ∇^F looks like taking a derivative while Berry's connection is usually written as a collection of functions. To say that ∇^F is Berry's connection, we need to represent ∇^F using some functions. Thinking in this way comes with the idea of a connection form.

2.4.2 Representing connection: the connection forms

On a coordinate neighborhood with a smooth orthonormal frame $\{\psi_1, \ldots, \psi_m\}$ of F, any section s can be written as

$$s(\vec{k}) = \sum_{i=1}^{m} s^i(\vec{k}) \psi_i(\vec{k}). \tag{27}$$

By Eq. (23), if we know $\nabla^F_X \psi_j$, we know every $\nabla^F_X s$. Expanding $\nabla^F_X \psi_j$ using $\{\psi_1, ..., \psi_m\}$ gives us

$$\nabla^F_X \psi_j = \sum_{i=1}^{m} \omega^i_j(X) \psi_i \tag{28}$$

Note $\nabla^F_{fX} s = f \nabla^F_X s$ for any smooth function f on M. Hence $\omega^i_j(fX) = f\omega^i_j(X)$. For any differential 1-form ω, we also have $\omega(fX) = f\omega(X)$. This suggests that $\{\omega^i_j\}^m_{i,j=1}$ are all differential 1-forms. These differential 1-forms are called the connection forms of the connection ∇^F.

2.4.3 Berry's connection is the induced connection ∇^F

Here we show Berry's connection is the induced connection ∇^F by computing the connection forms ω^i_j explicitly. For simplicity, we do this for the lower band of a two-band model. In this case, we only have ω^1_1.

$$(\nabla^F_{\frac{\partial}{\partial k^i}} \psi_1)(\vec{k}) = P(\vec{k})[\nabla_{\frac{\partial}{\partial k^i}} \psi_1(\vec{k})] = \langle \psi_1(\vec{k}), [\nabla_{\frac{\partial}{\partial k^i}} \psi_1](\vec{k}) \rangle \psi_1(\vec{k}) \tag{29}$$

Writing $\omega^1_1 = f_1 dk^1 + f_2 dk^2$ and using $dk^i(\frac{\partial}{\partial k^j}) = \delta^i_j$, we obtain

$$\omega^1_1 = \langle \psi_1, \nabla_{\frac{\partial}{\partial k^1}} \psi_1 \rangle dk^1 + \langle \psi_1, \nabla_{\frac{\partial}{\partial k^2}} \psi_1 \rangle dk^2 \tag{30}$$

This is the geometric connection form. We use the physics convention $A = i\omega^1_1$ for the real-valued Berry connection [5, 6]. The factor i matters when comparing curvature and Chern-number formulas. Note that the above computation of the connection forms of ∇^F generalizes easily to multiband systems: the resulting matrix $A = i\omega$ is the non-Abelian Berry connection [7]. These formulas are local: a nontrivial occupied bundle need not have a global smooth frame.

2.5 Berry's phase is the holonomy angle

As mentioned before, a connection is used to compare neighboring vectors. Therefore, we should be able to tell if the direction of a vector is changed when we move it around. Thinking in this way gives us parallel transport.

2.5.1 The geometric picture of parallel transport

To gain some geometric picture of parallel transport, we can again think of a 2D Riemannian manifold M with metric tensor g. The key point is that the tangent vector $\gamma'(t)$ of a geodesic γ does not change its length or direction under parallel transport. In Euclidean space, this is just the familiar picture of a constant vector along a straight line. For a general smooth curve α, we compare vectors at nearby points by transporting them along the short geodesic between those points, preserving length and oriented angle with the geodesic. Taking the infinitesimal limit gives a differential equation for $v(t) = \sum_i v^i(t) \frac{\partial}{\partial x^i}|_{\alpha(t)}$:

$$\frac{Dv(t)}{dt} \equiv \sum_i \frac{dv^i}{dt}(t) \frac{\partial}{\partial x^i}\bigg|_{\alpha(t)} + \sum_i v^i(t) \nabla_{\alpha'(t)} \frac{\partial}{\partial x^i} = 0. \tag{31}$$

Here ∇ is the connection discussed in Section 2.2.2, and each term is evaluated at $\alpha(t)$. The equation says precisely that v is parallel along α. For a general vector bundle, we define the covariant derivative along a curve in the same way, using a local frame in place of the coordinate tangent vectors.

2.5.2 Covariant derivative and parallel transport

Given a curve $\gamma : [0, T] \to BZ$ and a section $s(t) = \sum_{i=1}^m s^i(t) \psi_i(\gamma(t))$ along it, define

$$\begin{aligned} \frac{Ds}{dt} &\equiv \sum_i \frac{ds^i}{dt} \psi_i(\gamma(t)) + \sum_i s^i(t) \nabla^F_{\gamma'(t)} \psi_i \\ &= \sum_j \left(\frac{ds^j}{dt} + \sum_i \omega_i^j(\gamma'(t)) s^i(t) \right) \psi_j(\gamma(t)). \end{aligned} \tag{32}$$

Definition 2.3. *We say that $s(t)$ is parallel transported along γ if $\frac{Ds}{dt} = 0$ for all t.*

The local expressions agree on overlaps, so this definition does not depend on the chosen frame.

2.5.3 Adiabatic evolution is parallel transport

Now we explain in what sense adiabatic evolution is parallel transport [4–6]. Again, for simplicity, consider one isolated band of a two-band model. Suppose an external parameter changes slowly along a path $\vec{k}(t)$, so the instantaneous Hamiltonian is $H(\vec{k}(t))$. We set $\hbar = 1$ in this discussion. The Schrödinger equation is

$$i \frac{d\eta(t)}{dt} = H(\vec{k}(t)) \eta(t). \tag{33}$$

In the adiabatic limit, a state initially in the lower band has the leading-order form

$$\eta(t) \simeq e^{-i \int_0^t \lambda_1(\vec{k}(u))\, du} a(t) \psi_1(\vec{k}(t)). \tag{34}$$

The first factor is the dynamical phase. Substitute this expression into the Schrödinger equation and project onto the instantaneous eigenline. The dynamical terms cancel, giving

$$\frac{da}{dt} + a(t) \langle \psi_1, \nabla_{\vec{k}'(t)} \psi_1 \rangle = 0. \tag{35}$$

The projection is essential: the derivative of ψ_1 in the full bundle can have components outside the occupied line. By the definition of the induced connection,

$$P(\vec{k}(t)) \nabla_{\vec{k}'(t)} \psi_1 = \nabla^F_{\vec{k}'(t)} \psi_1. \tag{36}$$

Thus Eq. (35) is exactly

$$\frac{D(a(t)\psi_1(\vec{k}(t)))}{dt} = 0. \tag{37}$$

After removing the dynamical phase, adiabatic evolution is parallel transport for the induced connection. This statement holds in the adiabatic limit; it is not an exact solution of a general finite-speed Schrödinger evolution. Writing $a(t) = e^{i\theta(t)}$ with $a(0) = 1$, we obtain

$$\theta(t) = \int_0^t A(\vec{k}'(u))\, du, \qquad A = i \langle \psi_1, \nabla \psi_1 \rangle. \tag{38}$$

The geometric phase accumulated around a closed path is Berry's phase.

2.5.4 Example: holonomy angle in $S^2 \subset \mathbb{R}^3$

Parallel transporting a vector v around a closed loop doesn't always give us v but in general a vector w of the same length as v pointing in a different direction. For example, take a vector v at the north pole, transport it down a meridian to the equator, eastward through a quarter circle, and back to the north pole along another meridian, obtaining a vector w. Then the angle from v to w (counterclockwise) is $\pi/2$. This angle is called the holonomy angle.

2.5.5 Berry's phase is the holonomy angle

Since $a(t)\psi_1(\vec{k}(t))$ is parallel along the path, returning to $\vec{k}(0)$ after time T need not return the vector to itself. If we choose a frame along the loop with $\psi_1(\vec{k}(T)) = \psi_1(\vec{k}(0))$, the transported vector is $e^{i\theta(T)}\psi_1(\vec{k}(0))$. Hence Berry's phase $\theta(T)$, defined modulo 2π, is the holonomy angle [6]. If the chosen frame does not agree at the endpoints, its endpoint phase must also be included. Under a local change of frame $\psi_1 \mapsto e^{i\chi}\psi_1$, we have $A \mapsto A - d\chi$, while the closed-loop holonomy is unchanged.

2.5.6 Berry's phase for a multiband system

Now consider the lower m bands of an n-band system, as in Section 2.1. For geometric parallel transport, write $s(t) = \sum_{j=1}^{m} a^j(t)\psi_j(\vec{k}(t))$ in a local orthonormal frame of the occupied bundle. Then

$$\frac{Ds}{dt} = 0 \quad \Longleftrightarrow \quad \frac{da^i}{dt} + \sum_{j=1}^{m} \omega_j^i(\vec{k}'(t))a^j(t) = 0. \tag{39}$$

With $a = (a^1, \ldots, a^m)^{\mathsf{T}}$ and $\omega_{ij} = \omega_j^i$, its solution is

$$a(T) = \mathcal{P} \exp\left(-\int_0^T \omega(\vec{k}'(u))\, du \right) a(0). \tag{40}$$

Here $\mathcal{P}$ denotes path ordering: matrices encountered later along the path act on the left. For a loop, one must also identify the endpoint frames. The resulting unitary matrix is the non-Abelian holonomy; a change of initial frame conjugates this matrix.

The relation to physical adiabatic evolution needs a little care. If all states in the chosen subspace have the same instantaneous energy, removing their common dynamical phase gives this parallel transport, as in Wilczek and Zee's construction [7]. For a general group of bands with different energies, the projected Schrödinger equation in an instantaneous eigenbasis is instead

$$i\frac{db^i}{dt} + i\sum_j \omega_j^i(\vec{k}'(t))b^j(t) = \lambda_i(\vec{k}(t))b^i(t). \tag{41}$$

Removing separate band energies does not simply remove these dynamical terms from the off-diagonal couplings: relative dynamical phases remain. Nevertheless, the bundle and its geometric parallel transport are well defined whenever the whole group is separated by a gap from the remaining bands [4].

2.6 Berry's curvature is the infinitesimal Berry's phase

Now we are finally ready for Berry's curvature.

2.6.1 Example: the curvature of $S^2 \subset \mathbb{R}^3$

If we are living on a 2-sphere and not allowed to go into the third dimension, how do we know that the 2-sphere is curved? Note we can in principle know if we are traveling on a geodesic: just check if you are going at constant speed along a locally length-minimizing path. With geodesics, we can also have the notion of parallel transport as mentioned in Section 2.5.1. To measure curvature intrinsically, we can parallel transport a vector around the positively oriented boundary ∂D of a disk D. The oriented angle between the initial and final vectors is

$$\delta\theta(\partial D) = \int_D K\, dS \pmod{2\pi}, \tag{42}$$

where K is the Gaussian curvature of the 2-sphere. We obtain the Gaussian curvature by measuring the holonomy angle around an infinitesimal loop.

2.6.2 Curvature as infinitesimal holonomy

As mentioned in Section 2.6.1, the holonomy around a small loop measures curvature. On a Riemannian manifold, use the Levi–Civita connection and define

$$R(X,Y) = \nabla_X \nabla_Y - \nabla_Y \nabla_X - \nabla_{[X,Y]}. \tag{43}$$

The Lie bracket $[X,Y] = XY - YX$ is again a vector field. The expression $R(X,Y)Z$ is linear over smooth functions in X, Y, and Z, so its value at a point depends only on the vectors at that point, rather than on how they are extended nearby [8, Sections 7.7–7.8]. For a small coordinate rectangle based at p, traversed first in the X direction and then in the Y direction, parallel transport has the leading behavior

$$U_{\partial D} = I - \epsilon\delta\, R_p(X,Y) + O(\epsilon^2\delta + \epsilon\delta^2), \tag{44}$$

where ϵ and δ are its small coordinate side lengths and X, Y are the corresponding coordinate vectors. Thus curvature is infinitesimal holonomy. On an oriented surface, for positively oriented orthonormal vectors X, Y, the Gaussian curvature is $K = \langle R(X,Y)Y, X\rangle$. Equation (44) then gives the small-loop rotation angle K area(D) to leading order. The tensor R is the Riemann curvature tensor, with components

$$R_{ijkl} = \left\langle R\left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}\right)\frac{\partial}{\partial x^k}, \frac{\partial}{\partial x^l}\right\rangle. \tag{45}$$

We define curvature on a vector bundle in exactly the same way. For vector fields X, Y on BZ and a section s of F, set

$$R^F(X,Y)s = \nabla^F_X \nabla^F_Y s - \nabla^F_Y \nabla^F_X s - \nabla^F_{[X,Y]} s. \tag{46}$$

In the following discussion, R denotes this curvature of ∇^F.

2.6.3 Representing curvature: the curvature forms

From the definition of R, we have $R(X,Y) = -R(Y,X)$. We can check that $R(fX, gY) = fgR(X,Y)$ for all smooth functions f, g on BZ. Hence, for the same reason as in Section 2.4.2, we can write

$$(R(X,Y)\psi_j)(\vec{k}) = \sum_i \Omega^i_j(X,Y)\psi_i(\vec{k}) \tag{47}$$

The Ω^i_j are locally defined differential 2-forms on BZ representing the curvature tensor. In physics, we obtain Berry's curvature from Berry's connection. This is always true: we can obtain the curvature forms from the connection forms. But before that, we need another operation on differential forms: the exterior derivative.

2.6.4 The exterior derivative

Let M be an n-dimensional manifold, $\phi = (x^1, ..., x^n)$ be a local coordinate system on M and

$$\omega = \sum_{i_1 < \ldots < i_k} f_{i_1,\ldots,i_k} dx^{i_1} \wedge \cdots \wedge dx^{i_k} \tag{48}$$

be a differential k-form. Then the exterior derivative of ω is defined as

$$d\omega = \sum_{i_1 < \ldots < i_k} \sum_{j=1}^{n} \frac{\partial f_{i_1,\ldots,i_k}}{\partial x^j} dx^j \wedge dx^{i_1} \wedge \cdots \wedge dx^{i_k} \tag{49}$$

In particular, if $\omega = f$ is a 0-form, $df = \sum_{j=1}^n \frac{\partial f}{\partial x^j} dx^j$, which recovers our usual notion of differentiation.

2.6.5 Relating curvature and connection: the second structural equation

Proposition 2.2.

$$\Omega^i_j = d\omega^i_j + \sum_k \omega^i_k \wedge \omega^k_j \tag{50}$$

The proof of this identity involves some nontrivial computation and we will skip the proof here. We can view Ω^i_j and ω^i_j as the (i,j) entries of the matrices $\Omega = \sum_{i,j} \Omega^i_j e^i_j$ and $\omega = \sum_{i,j} \omega^i_j e^i_j$ (where e^i_j is the matrix whose (i,j) element is 1 and all other entries are 0) and define the bracket

$$[\omega, \omega] \equiv \sum_{i,j}\sum_{k,l} \omega^i_j \wedge \omega^k_l [e^i_j, e^k_l]. \tag{51}$$

Using $\omega_k^i \wedge \omega_j^k = -\omega_j^k \wedge \omega_k^i$, Eq. (50) can be rewritten as

$$\Omega = d\omega + \frac{1}{2}[\omega, \omega]. \tag{52}$$

Eq. (52) is called the second structural equation [8, Section 11.1].

2.6.6 Berry's curvature is the curvature of the induced connection

Again, consider the lower band of a two-band model. Since ω_1^1 is a 1-form, $\omega_1^1 \wedge \omega_1^1 = 0$. Write $\omega_j = \langle \psi_1, \nabla_{\partial/\partial k^j} \psi_1 \rangle$ for its two coefficient functions. Then

$$\Omega_1^1 = d\omega_1^1 = \left(\frac{\partial \omega_2}{\partial k^1} - \frac{\partial \omega_1}{\partial k^2} \right) dk^1 \wedge dk^2. \tag{53}$$

In our physics convention $A = i\omega_1^1$, the real-valued Berry curvature is $\mathcal{B} = dA = i\Omega_1^1$. For a loop bounding a disk on which a smooth frame is chosen, Stokes' theorem gives $\theta(\partial D) = \int_D \mathcal{B}$ modulo 2π. This is the precise sense in which Berry's curvature is infinitesimal Berry's phase. For multiple bands, the corresponding matrix-valued curvature is

$$\mathcal{B} = dA - iA \wedge A = i\Omega, \qquad A = i\omega. \tag{54}$$

These conventions will be used in the Chern-number formulas below.

3 Chern numbers

3.1 de Rham cohomology

Chern numbers are obtained by integrating differential forms representing Chern classes. To explain this construction, we first introduce de Rham cohomology [1, 9].

3.1.1 Closed forms and exact forms

Definition 3.1. *Let ω be a differential k-form on an n-dimensional manifold M. ω is called a closed form if $d\omega = 0$. ω is called an exact form if $\omega = d\eta$ for some differential $(k-1)$-form η.*

The following proposition tells us that exact forms are closed.

Proposition 3.1. *Let ω be a k-form on M, then $d(d\omega) = 0$.*

3.1.2 The de Rham cohomology groups

If ω and η are closed (respectively, exact) k-forms, then $a\omega + b\eta$ is also closed (respectively, exact) for any $a, b \in \mathbb{C}$. Hence closed k-forms form a vector space $Z^k(M)$, and exact k-forms form a subspace $B^k(M) \subset Z^k(M)$ by Proposition 3.1. For $k = 0$, set $B^0(M) = 0$. The de Rham cohomology group is the quotient space

$$H^k_{\mathrm{dR}}(M; \mathbb{C}) = Z^k(M)/B^k(M), \qquad k = 0, 1, \ldots, n. \tag{55}$$

Its dimension is a topological invariant called the kth Betti number. Roughly speaking, it counts independent k-dimensional cycles that are not boundaries. For example, the two independent directions around a torus give its first Betti number $b_1 = 2$.

3.2 The Chern class

With de Rham cohomology, we can construct the Chern class.

3.2.1 Constructing Chern classes

Let Ω be the curvature matrix of a Hermitian connection on the rank-m bundle F over M. The Chern forms $c_r(\nabla^F)$ are defined by expanding

$$\det\left(I_m + \frac{i}{2\pi}\Omega \right) = 1 + c_1(\nabla^F) + \cdots + c_m(\nabla^F). \tag{56}$$

When computing the determinant, replace the usual multiplication by wedge products. This makes sense because the entries of Ω are 2-forms, whose wedge products commute. The form $c_r(\nabla^F)$ has degree $2r$; in particular,

$$c_1(\nabla^F) = \frac{i}{2\pi} \operatorname{Tr} \Omega = \frac{1}{2\pi} \operatorname{Tr} \mathcal{B}. \tag{57}$$

Although Ω is written in a local frame, it transforms by conjugation under a change of frame. Thus the coefficients of Eq. (56) define global forms on M.

3.2.2 Chern classes are topological invariants

The Chern–Weil theorem says that these forms are closed and their cohomology classes are independent of the chosen connection [8, 9]. We define

$$c_r(F) = [c_r(\nabla^F)] \in H^{2r}_{\mathrm{dR}}(M;\mathbb{C}). \tag{58}$$

More precisely, this is the de Rham image of the integral Chern class in $H^{2r}(M;\mathbb{Z})$. This integral origin is what guarantees integer Chern numbers. If F and G are isomorphic bundles over M, their corresponding Chern forms differ by exact forms, so they represent the same cohomology classes. In this sense, Chern classes are topological invariants of the bundle, although their differential-form representatives depend on the connection.

3.3 Chern numbers from Chern classes

We obtain Chern numbers by integrating Chern classes.

3.3.1 Integration of differential forms

Let M be an oriented n-dimensional manifold and ω a compactly supported n-form. Compact support is automatic when M is compact. If the support lies in a positively oriented coordinate chart $\phi : U \to \phi(U) \subset \mathbb{R}^n$ and

$$\omega = f(x^1,\dots,x^n)dx^1 \wedge \cdots \wedge dx^n, \tag{59}$$

we define

$$\int_M \omega = \int_{\phi(U)} f(x^1,\dots,x^n)\, dx^1 \cdots dx^n. \tag{60}$$

For the general case, choose consistently oriented charts U_i and a smooth partition of unity $\{\rho_i\}$ subordinate to them: $\rho_i \geq 0$, each ρ_i is supported in U_i, and $\sum_i \rho_i = 1$. Then define

$$\int_M \omega = \sum_i \int_{U_i} \rho_i \omega. \tag{61}$$

The weights prevent overlapping charts from being counted twice. The result is independent of the charts and partition of unity [1].

3.3.2 Chern numbers are obtained by integrating Chern classes

Suppose M is compact, oriented, and without boundary, with even dimension n. Pick positive integers $r_1,\dots,r_l$ such that $2\sum_{j=1}^{l} r_j = n$. The corresponding Chern number is

$$\int_M c_{r_1}(\nabla^F) \wedge \cdots \wedge c_{r_l}(\nabla^F). \tag{62}$$

Because the Chern classes are integral, this number is an integer [9]. It does not depend on the connection: changing the representatives changes the integrand by an exact form, whose integral vanishes by Stokes' theorem. We cannot integrate a product of Chern forms over an odd-dimensional manifold to obtain a top-degree Chern number. However, we can integrate them over even-dimensional cycles inside it. For example, a 3D Brillouin torus has two-dimensional subtori on which first Chern numbers can be defined.

3.3.3 Computing the Chern number for a 2D periodic system

For a 2D system, the Brillouin zone is two-dimensional. The only potentially nonzero positive-degree Chern class is $c_1(F)$, and its integral gives the bulk Chern number

$$C = \frac{i}{2\pi}\int_{BZ} \operatorname{Tr}\Omega = \frac{1}{2\pi}\int_{BZ} \operatorname{Tr}\mathcal{B} \in \mathbb{Z}. \tag{63}$$

We orient BZ by $dk^1 \wedge dk^2$. For the lower band of a two-band model, Ω is a 1×1 matrix, so

$$\det\left(1 + \frac{i}{2\pi}\Omega\right) = 1 + \frac{i}{2\pi}\Omega. \tag{64}$$

Using Eq. (53), we obtain

$$C = \frac{i}{2\pi}\int_{BZ}\left(\frac{\partial\langle\psi_1, \nabla_{\partial/\partial k^2}\psi_1\rangle}{\partial k^1} - \frac{\partial\langle\psi_1, \nabla_{\partial/\partial k^1}\psi_1\rangle}{\partial k^2}\right) dk^1 dk^2. \tag{65}$$

This is the usual Berry-curvature formula. When $C \neq 0$, the eigenvector ψ_1 cannot be chosen smoothly and periodically over all of BZ: the local curvature expressions must be patched together. There is therefore no contradiction with Stokes' theorem, even though locally $\Omega = d\omega$. The same obstruction applies to a global occupied frame in the multiband case.

For a non-interacting 2D fermion system with its Fermi energy in a gap, C determines the quantized Hall response in units of e^2/h, with the sign depending on the convention for current and orientation [10]. A gapped band system with nonzero C is called a Chern insulator; Haldane's model shows that this can occur without a net magnetic flux through a unit cell [11]. We now ask how this bulk invariant appears at a physical edge.

4 Bulk–edge correspondence

So far, we have obtained an integer from the occupied bundle over the Brillouin zone. What does this integer tell us about a sample with a boundary? Here we explain how the first Chern number determines the net number of propagating edge channels. We will also sketch why an index theorem relates these two quantities.

4.1 From the bulk Hamiltonian to edge states

Consider the two-dimensional, non-interacting fermionic system in Section 2.1, with the Fermi energy at zero. We assume finite-range hopping and a bulk energy gap containing zero. No symmetry other than particle-number conservation is required; this is the setting usually called class A. Write

$$C = \int_{BZ} c_1(\nabla^F) = \frac{i}{2\pi} \int_{BZ} \operatorname{tr} \Omega. \tag{66}$$

Choose dimensionless crystal coordinates (k^1, k^2) of period 2π, orient BZ by $dk^1 \wedge dk^2$, and use Bloch waves proportional to $e^{i(k^1 x + k^2 y)}$.

Now keep only the sites with $x \geq 0$. We may change the Hamiltonian within finitely many layers of this boundary, while retaining translation symmetry in the y direction. The momentum k^2 is still a good quantum number. Fourier transformation along the boundary gives a family

$$H_{\text{edge}}(k^2) : \ell^2(\mathbb{N}_0) \otimes \mathbb{C}^n \longrightarrow \ell^2(\mathbb{N}_0) \otimes \mathbb{C}^n. \tag{67}$$

Here $\ell^2(\mathbb{N}_0)$ consists of square-summable wave functions in the direction perpendicular to the edge. Although the bulk has no states at zero energy, $H_{\text{edge}}(k^2)$ may have discrete eigenvalues $E_a(k^2)$ inside the bulk gap. The corresponding wave functions decay into the sample and are called edge states.

4.1.1 Counting edge states with a sign: spectral flow

Suppose first that the crossings of zero energy are isolated and have nonzero slope. Define

$$N_{\text{edge}} = \sum_{a, k_* :\, E_a(k_*) = 0} \operatorname{sgn}\left(\frac{dE_a}{dk^2}(k_*) \right), \tag{68}$$

where k^2 traverses the circle once in the increasing direction. This signed count is called the spectral flow through zero. An upward crossing contributes $+1$, and a downward crossing contributes -1. Degenerate crossings can be handled by a small perturbation or by the general definition of spectral flow; the resulting integer does not depend on this perturbation.

The group velocity along the edge is proportional to dE_a/dk^2. Thus N_{edge} counts channels moving in the positive y direction minus channels moving in the negative direction. It does not count all edge states without signs. Also, an edge branch need not remain an isolated eigenvalue for the entire circle: it can enter or leave the bulk bands. This explains how a periodic family can have nonzero spectral flow. A finite-dimensional periodic matrix family would have zero net spectral flow.

Theorem 4.1 (Bulk–edge correspondence)**.** *Under the assumptions above, and with the stated orientation and Fourier convention,*

$$N_{\text{edge}} = C. \tag{69}$$

Reversing the direction used to count edge propagation changes the sign. Keeping the same k^2 direction but retaining the opposite half-plane also changes the sign. The correspondence was established in the quantum Hall setting by Hatsugai; rigorous operator-theoretic formulations include those of Kellendonk, Richter and Schulz-Baldes, and Graf and Porta [12–14].

4.2 What is an index theorem?

Before sketching the relation to geometry, we introduce the index of an operator.

Definition 4.1. *A bounded operator $T : \mathcal{H}_1 \to \mathcal{H}_2$ between Hilbert spaces is Fredholm if its range is closed and both* $\ker T$ *and* $\ker T^*$ *are finite-dimensional. Its index is*

$$\operatorname{ind} T = \dim \ker T - \dim \ker T^*. \tag{70}$$

The same definition applies to suitable closed differential operators with specified domains. The index is unchanged by continuous deformations through Fredholm operators and by compact perturbations. Here "compact" means approximable in operator norm by operators of finite rank. An index theorem computes this analytic integer from topological data.

4.2.1 The Chern number as a Dirac index

There is a direct geometric example on our Brillouin torus. Equip the torus with its flat metric and a spin structure, and couple its Dirac operator to the occupied bundle F using the Berry connection. This produces a twisted Dirac operator D_F with two chiral parts, D_F^+ and $D_F^- = (D_F^+)^*$. The Atiyah–Singer index theorem gives [15]

$$\operatorname{ind} D_F^+ = \int_{BZ} \widehat{A}(TBZ)\,\operatorname{ch}(F) = \frac{i}{2\pi}\int_{BZ} \operatorname{tr}\Omega = C. \tag{71}$$

Only the two-form part is integrated. On the flat torus $\widehat{A}(TBZ) = 1$, and the two-form part of the Chern character $\operatorname{ch}(F) = \operatorname{tr}\exp(i\Omega/2\pi)$ is $c_1(\nabla^F)$. Thus a curvature integral equals a difference of numbers of zero modes.

This Dirac operator lives on momentum space and is an auxiliary mathematical construction. Its zero modes are not automatically the physical edge states. To obtain Eq. (69), we still need a theorem connecting the bulk bundle to the Hamiltonian on a half-plane.

4.3 How the boundary index theorem is applied

For a lattice system, this connecting result is naturally expressed using Toeplitz operators. These arise by taking an operator defined on the whole lattice and restricting it to a half-lattice. Restriction preserves multiplication only up to terms localized near the boundary. This small failure of multiplication is precisely where the boundary index enters. We describe the main steps without developing the full theory of operator algebras.

4.3.1 An edge unitary constructed from the Hamiltonian

Choose $\delta > 0$ so that $[-\delta, \delta]$ lies inside the bulk gap. Let $f(E)$ be a smooth nondecreasing function, equal to zero for $E \le -\delta$ and to one for $E \ge \delta$. Applying a function to a Hamiltonian means applying it to each energy in its spectral decomposition. Define

$$U(k^2) = \exp\bigl(2\pi i f(H_{\text{edge}}(k^2))\bigr). \tag{72}$$

For the bulk Hamiltonian, $f(H) = I - P$, where P is the occupied projection, so $\exp(2\pi i f(H)) = I$. For the half-plane Hamiltonian, however, edge energies inside the gap can make U different from I. Thus $U - I$ retains only information near the boundary. In the finite-range setting above, it is trace class on each perpendicular half-line and varies smoothly in trace norm, so we can form the winding number

$$W(U) = \frac{1}{2\pi i}\int_0^{2\pi} \operatorname{Tr}\left(U^{-1}\frac{dU}{dk^2}\right) dk^2. \tag{73}$$

Here Tr includes both the internal degrees of freedom and the perpendicular lattice sites.

An edge eigenvalue E_a contributes the phase $e^{2\pi i f(E_a)}$. As it crosses the gap upward, this phase makes one positive turn; a downward crossing gives one negative turn. Equivalently, its contribution to the integrand is $f'(E_a)\,dE_a/dk^2$. Integrating counts the signed crossings, so

$$W(U) = N_{\text{edge}}. \tag{74}$$

Changing the shape of f does not change this integer.

4.3.2 From winding to a Fredholm index, and back to the bulk

To see an actual Fredholm index, let P_+ act on $L^2(S^1;\ell^2(\mathbb{N}_0)\otimes\mathbb{C}^n)$ by projecting onto the Fourier modes e^{imk^2} with $m \geq 0$. Multiplication by U followed by this projection gives the Toeplitz operator

$$T_U = \mathsf{P}_+ U \mathsf{P}_+\big|_{\operatorname{Ran}\mathsf{P}_+}. \tag{75}$$

This is a second, mathematical compression, used to measure the winding along the edge. The Toeplitz index theorem relates the index to minus the winding; in our notation it gives [16]

$$\operatorname{ind} T_{U^*} = W(U). \tag{76}$$

The assumptions on $U - I$ ensure that these Toeplitz operators are Fredholm.

The remaining ingredient is the boundary index theorem for the Toeplitz extension of the bulk lattice algebra [13]. Its relevant statement, with our conventions, is

$$\operatorname{ind} T_{U^*} = \frac{i}{2\pi}\int_{BZ} \operatorname{tr}\Omega. \tag{77}$$

The operator $\widetilde{P} = I - f(H_{\text{edge}})$ has P as its bulk counterpart, but need not itself be a projection. Exponentiating gives $\exp(2\pi i\widetilde{P}) = U^*$. This is the boundary construction behind the theorem, and it explains why the adjoint U^* appears in Eq. (77). The theorem says that the Chern-number pairing before this operation agrees with the winding-index pairing afterwards. This is the nontrivial mathematical step: one cannot obtain it merely by applying Stokes' theorem to Berry curvature. The boundary of the physical sample is not a boundary of the Brillouin torus.

Combining Eqs. (74)–(77) gives Eq. (69). The Atiyah–Singer theorem explains the geometric meaning of the bulk integer, while the Toeplitz boundary theorem identifies that integer with physical edge spectral flow.

4.4 Why are the edge channels robust?

Suppose $C = 1$. There may be one positive-velocity crossing, or several positive- and negative-velocity crossings, but their signed difference must be one. Changing the boundary can create or remove opposite-moving pairs without changing this difference. Consequently, local changes of the edge cannot remove all crossings while the bulk gap remains open. If $C = 0$, edge states may still occur, but this Chern-number argument does not protect a nonzero net chirality. On a wide strip the two edges have opposite signed counts when measured along the same coordinate; the half-plane formulation isolates one edge and avoids their finite-width hybridization.

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